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Mollification of singular dissipation models for order improvement in discrete element method simulations of granular media

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Abstract:	The Discrete Element Method (DEM) is a numerical technique widely used to simulate granular materials. It approximates the trajectory of each particle in a granular flow by integrating Newton's equations of motion with forces calculated from pairwise contacts between the grains. There exists a multitude of force schemes in the literature that model the force arising from binary collisions; among them, the Kuwabara-Kono force scheme is one of the most frequently used to model the dissipative force in a particle-particle interaction. Combining the Kuwabara-Kono force scheme with DEM, and performing the numerical integration with a Verlet-type method of order two does not yield second order convergence. Instead, the order of convergence has been mathematically proven to be 1.5 because such a force scheme makes the right-hand side of the equations of motion have an unbounded derivative at the beginning of each collision, i.e., it becomes non-Lipschitz. Many foundational theorems on numerical approximations for the Ordinary Differential Equation (ODE) $y'^{\prime} \text{ of } t = f \text{ of } t, y \text{ of } t$ include a hypothesis requiring f to be a Lipschitz continuous function. Without this assumption, the numerical approximation might not converge to the solution, or it may do so at a lower order than the theory predicts. In this work, we propose a regularization of the Kuwabara-Kono force model via mollification that removes this singularity, and we show numerically that the Verlet method combined with this regularized force model can integrate collisions with second order accuracy. We implement the regularized model for a collision dominated flow, and by proposing an appropriate metric, we verify the improvement in the error of the simulations when compared with the non-regularized model. The technique proposed in this work can, in fact, be used to regularize non-Lipschitz right-hand sides of many ODEs that needs to be numerically integrated with high order methods.



Universidade de Brasília

Departamento de Matemática

Brasília, Brazil, 6 April 2026.

Subject: Submission of manuscript to Computer Physics Communications.

Dear Editor,

I am enclosing herewith the manuscript entitled "***Mollification of singular dissipation models for order improvement in discrete element method simulations of granular media***" for consideration for publication in Computer Physics Communications (CPC).

This manuscript is the outcome of an original research that proposes a regularization of the well-known Kuwabara-Kono force model used in DEM simulations of granular materials via mollification. The Kuwabara-Kono model, in its original form, presents a singularity at the beginning of particle collisions that penalizes the order of convergence of leapfrog-Verlet methods commonly used to integrate the equations of motion. Our proposed regularization removes this singularity and restores the expected order 2 of convergence of the integration method.

As the relevance and usage of DEM solvers increase at a fast pace, please allow me to draw your attention to some recent works dealing with understanding and improving problems in the integration of the equations of motion in DEM simulations, some of which were published in CPC:

Improved velocity-Verlet algorithm for the discrete element method, ***Computer Physics Communications***, Volume 310, 109524, 2025, <https://doi.org/10.1016/j.cpc.2025.109524>;

Kuwabara-Kono numerical dissipation: a new method to simulate granular matter, IMA Journal of Applied Mathematics, Volume 85, 1, pp 27–66, 2020, <https://doi.org/10.1093/imamat/hxz034>;

SPIRAL: An efficient algorithm for the integration of the equation of rotational motion, ***Computer Physics Communications***, Volume 297, 109077, 2024, <https://doi.org/10.1016/j.cpc.2023.109077>.



Universidade de Brasília

Departamento de Matemática

Understanding the impact and potential consequences of the numerical integration of the equations of motion is essential for correctly pinpointing the underlying physics of the (granular) phenomena investigated. The results and the technique proposed in this work are of relevance to both users and developers of DEM solvers and, therefore, we believe that CPC is the most pertinent forum to submit our work to, as it is precisely directed to this broad audience.

The code used to generate the results presented in the manuscript can be found in the link below:

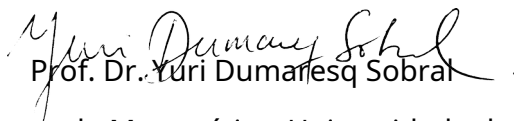
<https://gitfront.io/r/gnb1007/XE9o9uZr8YvD/Sandbox/>

If you deem appropriate, please do share it with the reviewers. We are happy to include this link in the manuscript at a later stage if appropriate.

With this submission, I would like to state that the above mentioned manuscript has not been published in its current form elsewhere nor is accepted for publication elsewhere. Furthermore, **this manuscript is not currently under consideration by any other journal and will not be submitted elsewhere during the review process** in CPC. In addition, I state that my co-author is fully aware of this submission.

Please do not hesitate to contact me in case you need further information regarding this submission or about the manuscript itself. May I take this opportunity to thank very much in advance for all your kind attention.

Yours sincerely,


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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Mollification of singular dissipation models for order improvement in discrete element method simulations of granular media

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Abstract

The Discrete Element Method (DEM) is a numerical technique widely used to simulate granular materials. It approximates the trajectory of each particle in a granular flow by integrating Newton's equations of motion with forces calculated from pairwise contacts between the grains. There exists a multitude of force schemes in the literature that model the force arising from binary collisions. Among them, the Kuwabara-Kono force scheme is one of the most frequently used to model the dissipative force in a particle-particle interaction. Combining the Kuwabara-Kono force scheme with DEM, and performing the numerical integration with a Verlet-type method of order two does not yield second order convergence. Instead, the order of convergence has been mathematically proven to be 1.5 because such a force scheme makes the right-hand side of the equations of motion have an unbounded derivative at the beginning of each collision, i.e., it becomes non-Lipschitz. Many foundational theorems on numerical approximations for the Ordinary Differential Equation (ODE) $y'(t) = f(t, y(t))$ include a hypothesis requiring f to be a Lipschitz continuous function. Without this assumption, the numerical approximation might not converge to the solution, or it may do so at a lower order than the theory predicts. In this work, we propose a regularization of the Kuwabara-Kono force model via mollification that removes this singularity, and we show numerically that the Verlet method combined with this regularized force model can integrate collisions with second order accuracy. We implement the regularized model for a collision dominated flow, and by proposing an appropriate metric, we verify the improvement in the error of the simulations when compared with the non-regularized model. The technique proposed in this work can, in fact, be used to regularize non-Lipschitz right-hand sides of many ODEs that need to be numerically integrated with high order methods.

Keywords— Granular materials, Discrete element method, Verlet method, Leapfrog method, Kuwabara-Kono model, Order of convergence, Mollifier

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1 Introduction

Granular materials are involved in many processes which directly affect humankind: the lay of the land is heavily influenced by erosion and sediment transport; rice — which accounts for more than 20% of the calories consumed worldwide [13] — behaves as a granular material during harvest, transport, storage, filtering and packaging; landslides, which claim the lives of thousands each year, are also a granular phenomenon. Partially because of this impact in the life of the general population, granular flows have been extensively studied through modeling and computational simulations [1].

One of the earliest such models designed to describe granular matter was introduced in [9]. The method is based on calculating the motion of each particle individually, with interactions governed by prescribed contact rules between pairs of particles. This strategy is closely related to molecular dynamics algorithms, which are widely applied in the simulation of atomic and molecular systems [27]. Over time, the approach of [9] became known as the Discrete Element Method (DEM), which, today, is one of the most broadly used computational techniques for studying granular flows across diverse contexts [14]. More recently, several adaptations of DEM have been proposed [24], and coupling with fluid flow has also been incorporated through different approaches [19, 29].

Mathematics Subject Classification (2020): 76T25, 70F40, 70F35, 70-10

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The core of DEM lies in the constitutive laws chosen to represent the interaction forces during particle contacts. For dry granular flows, two categories of forces are usually considered [25]: elastic forces, which account for the restoring action during contact, and dissipative forces, which model energy losses. Among the latter, friction — arising from tangential relative motion — is the most significant. Additional dissipative forces are associated with the normal component of the relative motion along the line connecting particle centers [25].

The simplest constitutive model for normal forces in DEM combines linear elastic and viscous terms [25]. Besides its straightforward formulation, this model allows closed-form derivations of several properties of particle contacts, which serve as guidelines for parameter selection in simulations. However, such basic representation is insufficient to fully describe inelastic collisions [25, 26].

Numerous other models have been developed [25, 26], each offering different advantages and limitations. Among them, the dissipative Hertz model introduced in [21] — and later rigorously derived from material considerations in [3] — has shown some of the best agreement with experimental evidence. This nonlinear force law, which combines Hertzian elasticity [16] with a nonlinear dissipative contribution, forms the foundation of the present work. The derivation in [3] clarified several aspects not present in [21], such as validity conditions, the explicit form of the dissipative constant in terms of material parameters, and a physical interpretation of this constant. Nevertheless, in the literature this formulation is most commonly referred to as the Kuwabara–Kono force model.

The equations of motion for the particles, which are obtained from Newton’s laws, are typically integrated with some Verlet-like scheme [15]. Other techniques for numerical approximation have been used, such as symplectic [15, 27, 30], Runge–Kutta [22], Euler, and implicit methods [7]. However, few, if any, researchers have concerned themselves with the compatibility between the integration method and the force scheme. This is the focus of the present text.

Historically, the mathematical foundations of DEM simulations of granular materials seem to have been neglected. This neglect can cause implementations to act in erroneous — often undetected — ways, which could lead to costly mistakes in industry, civil construction and preparation of natural disasters. As an example, it has recently been shown in [28] that most open source DEM simulators of granular materials can present unphysical behavior in polydisperse flows, which is caused by calculating the relative tangential velocity from quantities staggered by $\frac{\Delta t}{2}$: the velocities are evaluated at time $t + \frac{\Delta t}{2}$, while the unit tangential vector is calculated from positions at time $t + \Delta t$.

Another possible setback caused by not being mindful of interactions between integrator and equation is the loss of precision. It is well known that the proofs of convergence for most ODE integrators assume that the function on the right-hand side of the expression $y'(t) = f(y, t)$ is Lipschitz continuous [2]. Without this hypothesis, there is no mathematical guarantee of convergence, let alone convergence with a particular order. For studies where the trajectories of individual particles are tracked, such as [8, 10, 11], a reduction in order of convergence caused by irregular contact forces may lead to erroneous results or, at least, require the usage of smaller time step sizes, which requires more computational power.

An example of a contact force scheme that is not Lipschitz continuous is the so-called Kuwabara–Kono model, which introduces dissipation through a term proportional to the square root of particle overlap [21]. Consequently, its first derivative becomes singular at the onset of contact, when the overlap transitions through zero. In a previous work, we provided a formal proof that the leapfrog method applied to the ODE $y'' = y^q$, with $q \in (0, 1)$, is still convergent, but its order becomes $1 + q$, instead of the expected 2 [5]. A similar argument can be used to show that using a predictor-corrector leapfrog scheme [27] to the equations of motion of particles whose normal dissipation is based in Kuwabara–Kono method will result in order 1.5 [5, 6, 18]. Surprisingly, this phenomenon seems to have been overlooked, until recently, by the community of those working with granular materials via DEM.

To the authors’ knowledge, the first to recognize this order decrease were [18]. However, beyond identifying the effect, no theoretical explanation was offered in [18]; instead, the authors obtained a new equation by making a clever change of variables, which avoids any singularities in the derivative of the right-hand side; in fact, its right-hand side becomes Lipschitz continuous. This approach is able to successfully remedy the order defect phenomenon in the very specific case where the normal forces are a combination of a Hertz elastic force and the Kuwabara–Kono damping.

In this work, we propose an alternative solution to this order defect. Instead of transforming the specific equation of this problem, we introduce a new parameterized model, which approximates the desired force scheme arbitrarily well as its parameters go to zero. The advantages of this approach over the one discussed above are: although we focus on the Hertz–Kuwabara–Kono [21] case, the proposed solution can be easily adapted to any other non-Lipschitz force schemes. Furthermore, it is easily incorporated into the design of existing DEM software, which does not appear to be the case for the solution proposed in [18].

This work is structured as follows: in section 2, we recall the definitions of some relevant models and integration schemes and state more rigorously the problem we are trying to solve. Then we cover the prerequisites necessary for our proposed solution in section 3. The new model proper is introduced in section 4, where we also provide analyses of both the order and the physics of binary collisions under the new model. In section 5, we introduce a metric that measures, as a function of time, the maximum “separation” of a group of simulations, and then we compare the value of this metric for the regularized and standard models in the case of a granular explosion. Finally, section 6 summarizes the findings of this work and presents directions for future work.

2 Statement of the problem

2.1 The Hertz–Kuwabara–Kono force model

Consider two spherical particles, P_1 and P_2 , positioned near one another on a plane orthogonal to the gravitational direction, such that their interaction results in a strictly normal collision, i.e., a collision in which the contact forces act exclusively along

the line joining the particle centers. The dynamics of each particle P_i are then dictated by Newton's second law, involving only the normal contact forces, namely

$$m_i \mathbf{x}_i''(t) = \mathbf{F}_i(t), \quad (1)$$

where m_i denotes the particle mass, $\mathbf{x}_i(t)$ its position vector, and $\mathbf{F}_i(t)$ the corresponding normal contact force, for $i \in 1, 2$. Here, the primes in [equation \(1\)](#) represent derivatives taken with respect to time.

In DEM, the overlap between particles P_i and P_j is defined as

$$\xi_{i,j}(t) := \max\{0, R_i + R_j - \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|\}, \quad (2)$$

where R_i and R_j are the radii of P_i and P_j , respectively. Most contact forces laws used in DEM are expressed as functions of this overlap and its derivative. Although it is possible to model tangential forces in DEM [\[9\]](#), we will only discuss and use normal force models in this work.

Normal contact forces in DEM are typically a sum of two terms: one modelling the elastic response of the material and the other accounting for the damping observed in real life collisions. Among several options of elastic contact forces, the Hertz model is one of the most used [\[16\]](#), which is expressed as the product of a constant $k_{i,j} \in (0, \infty)$ — which depends on the material and radii of the particles — and $\xi_{i,j}(t)^{3/2}$. As for the damping term, one of the most commonly employed is the Kuwabara-Kono model [\[21\]](#), whose expression is $\gamma \xi_{i,j}(t)^{1/2} \xi'_{i,j}(t)^{1/2}$, where γ is the damping coefficient. We will refer to this combination by “Hertz-Kuwabara-Kono model”. Therefore, the normal contact force between particles P_i and P_j will be given by

$$\|\mathbf{F}_i(t)\| = k_{i,j} \xi(t)^{3/2} + \gamma \xi(t)^{1/2} \xi'(t). \quad (3)$$

The constants $k_{i,j}$ and γ can be determined in terms of geometric and material parameters of the particles involved [\[3\]](#). For most materials, these parameters are known within a small error due to the experimental nature of the measurement procedures. This model gained wide use because it provides a realistic description of normal collisions in granular systems [\[21, 25, 26\]](#).

2.2 The predictor-corrector leapfrog method

In DEM simulations of granular systems, the time integration of [equation \(1\)](#) is typically carried out using the Verlet algorithm [\[15\]](#). When the interaction forces are independent of the velocity $\mathbf{v}_i(t) = \mathbf{x}_i'(t)$ of particle P_i , this scheme is mathematically equivalent to the leapfrog method.

If the force law explicitly involves velocity, however, the standard Verlet formulation must be modified in order to maintain second-order accuracy [\[27\]](#). This is accomplished by employing a two-stage leapfrog approach. At the n -th step (where $n \in \mathbb{N}$), the first stage consists of finding a provisional position $\hat{\mathbf{x}}_{i,n+1}$ for particle P_i . Here, variables with an extra subindex represent the approximations found by the integration method, where the additional subindex multiplied by the time step size gives the time which the variable approximates the quantity, i.e. $\mathbf{x}_{i,n} \approx \mathbf{x}_i(nh)$, where $h \in (0, \infty)$ is the time step size. The velocity $\mathbf{v}_{i,n}$ is then reconstructed at time $t_n := nh$ by means of a second-order centered finite difference. Using this velocity estimate, the force $\mathbf{F}_{i,n}$ acting on the particle at t_n is evaluated. A second leapfrog step is then applied to update the particle's position with this corrected force. Restricting attention to one-dimensional motion along the line connecting particle centers, one-step of this integration scheme can be expressed as

$$\left\{ \begin{array}{l} \hat{\mathbf{v}}_{i,n+\frac{1}{2}} = \mathbf{v}_{i,n-\frac{1}{2}} + \frac{h}{m_i} \sum_{j=1}^M \mathbf{f}\left(t_n, \xi_{i,j,n}, \xi'_{i,j,n-\frac{1}{2}}\right); \\ \hat{\mathbf{x}}_{i,n+1} = \mathbf{x}_{i,n} + h \hat{\mathbf{v}}_{i,n+\frac{1}{2}}; \\ \mathbf{v}_{i,n} = \frac{\hat{\mathbf{x}}_{i,n+1} - \mathbf{x}_{i,n-1}}{2h}; \\ \mathbf{v}_{i,n+\frac{1}{2}} = \mathbf{v}_{i,n-\frac{1}{2}} + \frac{h}{m_i} \sum_{j=1}^M \mathbf{f}\left(t_n, \xi_{i,j,n}, \xi'_{i,j,n}\right); \\ \mathbf{x}_{i,n+1} = \mathbf{x}_{i,n} + h \mathbf{v}_{i,n+\frac{1}{2}}, \end{array} \right. \quad (4)$$

where $M \in \mathbb{N}$ is the number of particles in the system and $\mathbf{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a function whose output is the contact force.

2.3 The problem

As thoroughly discussed in [\[5\]](#), combining [equations \(3\) and \(4\)](#) results in a convergence of order 1.5. This can more easily be observed through a binary normal collision. For simplicity, we assemble this collision as a 2-dimensional simulation.

We start by having two particles, say P_1 and P_2 , each with density ρ and positioned with their centers along the x -axis such that the distance between their center is exactly the sum of their radii, i.e., they are at the imminence of contact. The particle P_2 starts at rest while P_1 has an initial velocity directed from its center to the center of P_2 and with some magnitude v_0 . The simulation is then allowed to run for a total physical time of $t_0 := 15 \times 2^{-27} \text{s} \approx 1.1 \times 10^{-11} \text{s}$. This procedure is then repeated several times for different values of the time step size h among a fixed set of time step sizes $\{h_1, h_2, \dots, h_k \in (0, \infty)\}$, and the x -coordinate of the final position of P_1 is recorded in each repetition, which we call x_i , for $i \in \mathbb{N} \cap [1, k]$. It is important that

each of the h_i divides t_0 with no remainder, to ensure that all simulations end at the same moment in time. If we let h_k be the smallest time step size used in this procedure, we can define the error of x_i , $i \neq k$, with respect to x_k as

$$\Delta x_i := |x_i - x_k|. \quad (5)$$

By fitting a power law to the set $\{(h_i, x_i) \in \mathbb{R}^2 \mid i \in \mathbb{N} \cap [1, k]\}$, we can find the order of the integration method applied to the force scheme.

We performed the order-measuring procedure described above with the parameters given in [table 1](#) twice; the only difference between them was that, in [equation \(3\)](#), we set $\gamma = 0$ for one of them. The results are shown in [figure 2](#), which clearly reproduces the order difference presented in [\[5, 18\]](#).

Table 1: Values of the material and model parameters used to simulate a binary normal collision between two particles.

Particle	Model	Initial Condition
$\rho = 1308.57 \text{kg/m}^3$	$k_{1,2} \approx 1.27 \times 10^9 \text{N/m}^{1.5}$	$v_0 = 0.5 \text{m/s}$
$r = 0.5 \text{m}$	$\gamma = 190 \text{kg}/(\text{m}^{0.5} \text{s})$	

As proved in [\[5\]](#), this difference is caused by the derivative of the Kuwabara-Kono term being unbounded as $\xi \rightarrow 0$. Our goal in this work is to create a family of smooth, numerically implementable models parameterized by a value ε that approximates the right-hand side of [equation \(3\)](#) arbitrarily well as $\varepsilon \rightarrow 0$. Formally, we wish to find, for each $\varepsilon \in (0, \infty)$, a smooth function $f_\varepsilon : [0, \infty)^3 \times \mathbb{R} \rightarrow [0, \infty)$ such that

$$\lim_{\varepsilon \rightarrow 0} f_\varepsilon(k, \gamma, x, v) = kx^{3/2} + \gamma x^{1/2}v, \quad (6)$$

for every $k, \gamma, x \in [0, \infty)$ and $v \in \mathbb{R}$.

3 Prerequisites

3.1 Mollifiers

Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$\phi(x) := \begin{cases} \frac{1}{C} \exp\left(\frac{1}{x^2 - 1}\right) & \text{if } -1 < x < 1; \\ 0 & \text{otherwise,} \end{cases} \quad (7)$$

where

$$C := \int_{-1}^1 \exp\left(\frac{1}{x^2 - 1}\right) dx \approx 0.444 \quad (8)$$

is chosen such that the integral of $\phi(x)$ equals 1. The function defined by [equation \(7\)](#) is called the “standard mollifier” and is depicted in [figure 1](#). The interested reader can find more details as well as several properties of mollifiers in [\[12\]](#).

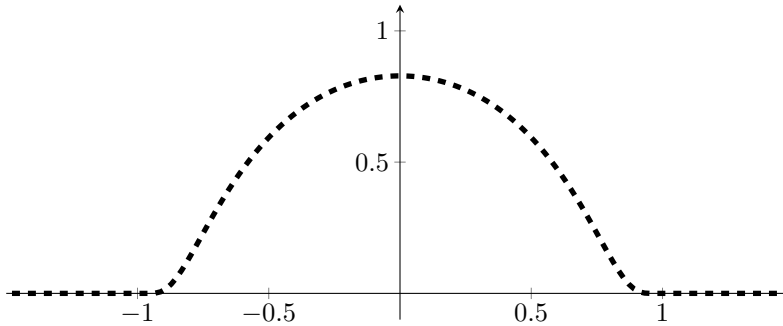


Figure 1: The standard mollifier.

For any $\varepsilon \in (0, \infty)$, let $\phi_\varepsilon : \mathbb{R} \rightarrow \mathbb{R}$ define the rescaled mollifier

$$\phi_\varepsilon(x) := \frac{1}{\varepsilon} \phi\left(\frac{x}{\varepsilon}\right). \quad (9)$$

Given $f : \mathbb{R} \rightarrow \mathbb{R}$ a locally integrable function, one can define its ε -mollification as the convolution of ϕ_ε and f . This convolution yields a new real function, denoted by $\phi_\varepsilon * f$, which can be computed via the expression

$$(\phi_\varepsilon * f)(x) = \int_{-\varepsilon}^{\varepsilon} \phi_\varepsilon(z) f(x - z) dz. \quad (10)$$

The new function originated by [equation \(10\)](#) has the desired property of being infinitely differentiable in $\mathbb{R}$, while approximating the original function f arbitrarily well as $\varepsilon \rightarrow 0$. This is formally stated in the result below.

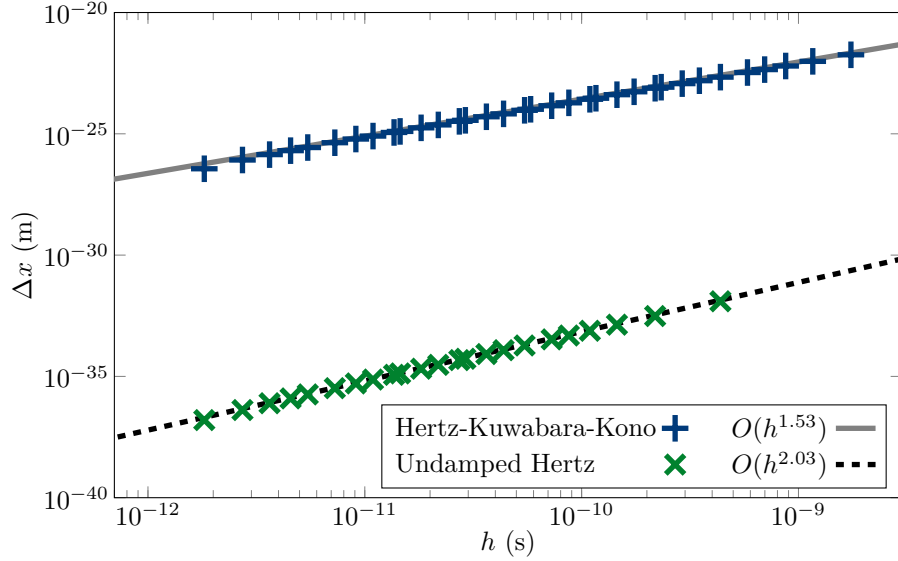


Figure 2: Results¹ of the order-measuring procedure discussed in subsection 2.3. The values of the parameters can be found in table 1. Hertz model without damping means that $\gamma = 0$ in equation (3).

Theorem 3.1. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ a locally integrable function. Then, the following statements are true:*

- (1) $\phi_\varepsilon * f$ is infinitely differentiable;
- (2) $\phi_\varepsilon * f \rightarrow f$ almost everywhere as $\varepsilon \rightarrow 0$;
- (3) If f is continuous, then $\phi_\varepsilon * f \rightarrow f$ uniformly on compact subsets of $\mathbb{R}$ as $\varepsilon \rightarrow 0$.

Proof. See [12]. □

In other words, $\phi_\varepsilon * f$ is an infinitely differentiable approximation of f . How well it approximates f depends on how small ε is taken.

3.2 Extended square root and ε -shift

In order to properly calculate the integral in equation (10) for non-negative values of x , the function f must be defined on $[-\varepsilon, \infty)$. In the case of the square root function, which is the one appearing in the dissipation term of equation (3), this is not true. For this reason, we continuously extend the ordinary square root function to negative numbers by making $\sqrt{x} = 0$ if $x < 0$. This extension is denoted by $\sqrt{\cdot} : \mathbb{R} \rightarrow \mathbb{R}$ and will substitute the traditional square root function from now on.

If we use the extended square root function in the place of f in equation (10) we observe that the ε -mollification of $\sqrt{\cdot}$ is not zero when $x = 0$, as illustrated in figure 3(a). In fact, notice that the integrand of equation (10) becomes $\phi_\varepsilon(z) \sqrt{-z}$ for $x = 0$. Then, if $z \in (-\varepsilon, 0)$, the ε -mollifier $\phi_\varepsilon(z)$ and $\sqrt{-z}$ are non-zero. Since both functions are strictly positive when they are not zero, this results in a positive value for the overall convolution. This could pose a problem for the force model, since it would mean that a non-zero normal force would exist between two particles which are not in contact. A way to prevent this is to right-shift the function $\sqrt{\cdot}$ by ε . That is, for each $\varepsilon \in (0, \infty)$, define the right-shift function $\tau_\varepsilon : \mathbb{R} \rightarrow \mathbb{R}$ as

$$\tau_\varepsilon(x) = x - \varepsilon. \quad (11)$$

Then, the ε -mollification of $\sqrt{\cdot} \circ \tau_\varepsilon$, which is depicted in figure 3(b), is always zero when $x \leq 0$, i.e. when there is no contact between the particles. For convenience of notation, from now on we define

$$_\varepsilon\sqrt{\cdot} := \phi_\varepsilon * (\sqrt{\cdot} \circ \tau_\varepsilon). \quad (12)$$

3.3 Efficiently computing $_\varepsilon\sqrt{\cdot}$

Since the expression for the value of an ε -mollification is given by an integral on a limited domain (see equation (10)), one can approximate its value by any of many integration methods available in the literature [2]. However, naively applying any of these methods to equation (10) yields approximations which become very inaccurate near 0, as shown in figure 4(a). These inaccuracies present themselves in two forms: a “delay” before the approximation becomes non-zero and a non-smooth behaviour in some points. These inaccuracies are caused by the finite number of samples each integration method uses to approximate equation (10). This can be better understood by the schematic plots presented in figure 5(a).

¹Some readers may find odd that the Δx axis spans to 10^{-40} , given that the rounding machine epsilon for the IEEE standard for double precision floating points is $\approx 1.11 \times 10^{-16}$. To reach these values, the authors implemented (in software) and used the “double double” arithmetic type [4, 17, 20], which provides increased precision at a (relatively) low computational cost.

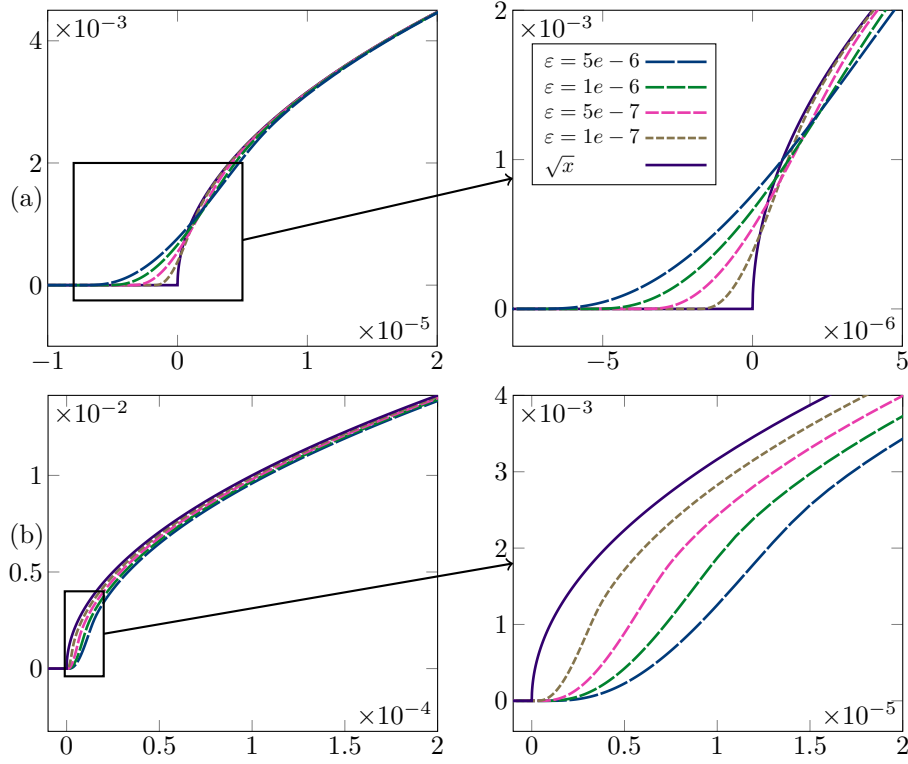


Figure 3: (a) Comparison between $\sqrt{\cdot}$ and the ε -mollification of $\sqrt{\cdot}$ for different values of ε . (b) Comparison between $\sqrt{\cdot}$ and the ε -mollification of $(\sqrt{\cdot} \circ \tau_\varepsilon)$, ${}_\varepsilon\sqrt{\cdot}$, for different values of ε . All curves in (b) go through (0, 0). In both figures, mollifications were computed using a composite midpoint rule with 1000 sub-intervals.

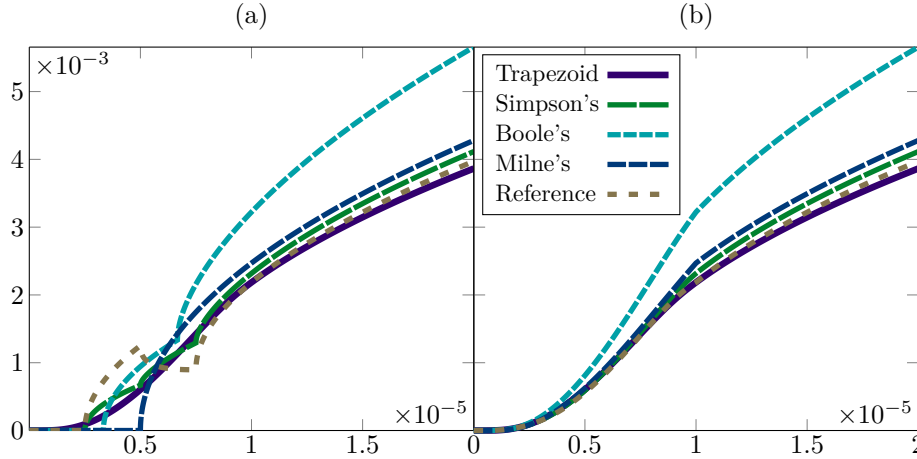


Figure 4: (a) Comparison of the approximations of ${}_\varepsilon\sqrt{x}$, for $\varepsilon = 5 \times 10^{-6}$, yielded by different integration methods. (b) Same comparisons, but the integration domain was changed at each x so as to match the support of the integrand. In either plot, the same reference curve was used and it was computed using a composite midpoint rule with 1000 sub-intervals.

Given $x_0 \in [0, \infty)$, the factors of the integrands of the convolution in ${}_\varepsilon\sqrt{x_0}$ take the form of $\phi_\varepsilon(z)$ and $\sqrt{(x_0 - z - \varepsilon)}$. This latter factor is just the square root function right-shifted by $(\varepsilon - x_0)$ and then mirrored about the vertical axis. This is illustrated in the colored curves of figure 5(a). When computing ${}_\varepsilon\sqrt{x_0}$ with a numerical integration method, the aforementioned inaccuracies arise as the value x_0 crosses a sampling point of the integration method. This causes the approximation to be calculated with more samples, which changes its behavior. For instance, in figure 5(a), the yellow curve indicates that the integral is calculated with only one sampling of the function while for the blue curve the integral is calculated with two points.

One can also observe from figure 5(a) that the support of the product of the integrands of ${}_\varepsilon\sqrt{x}$ starts as the empty set when $x = 0$ and increase with x until it becomes the whole interval $[-\varepsilon, \varepsilon]$ when $x = 2\varepsilon$. This suggests that one can improve the approximations of ${}_\varepsilon\sqrt{x}$ near $x = 0$ by applying the integration method on the support of the integrand, instead of the entire interval $[-\varepsilon, \varepsilon]$, as shown in figure 5(b). It can be readily seen that this support is $[-\varepsilon, \min\{x - \varepsilon, \varepsilon\}]$. The results of using

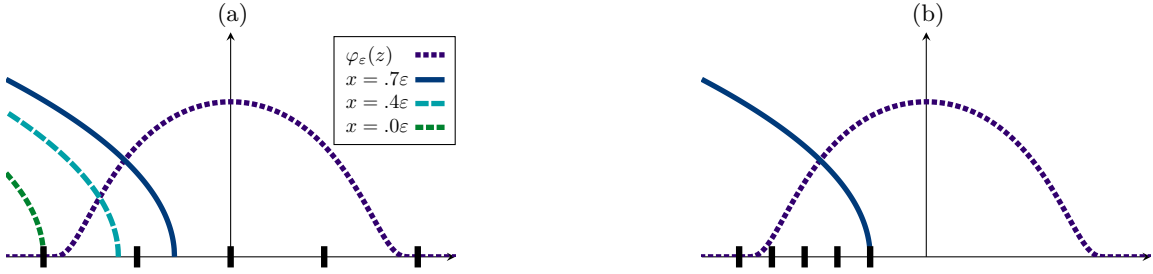


Figure 5: Schematic representation of the factors in the integrand of $(\varepsilon\sqrt{\cdot})(x)$ and the sampling points of the numerical integration method. In order to keep the notation consistent with [equation \(10\)](#), the horizontal axis represents the z values, while x are constants. Furthermore, the lines whose key is of the form $x = q\varepsilon$, for some $q \in [0, 1]$, are plots of $f(q\varepsilon - z)$ for $f = \sqrt{\cdot} \circ \tau_\varepsilon$, i.e., they are given by $\sqrt{q\varepsilon - z - \varepsilon} = \sqrt{(q-1)\varepsilon - z}$. Note how the support of the product of both functions changes as the point where the convolution is calculated changes. (a) The notches on the z -axis are the sampled points of the numerical integration method applied to the interval $[-\varepsilon, \varepsilon]$. (b) The notches on the z -axis are the sampled points of the numerical integration method applied only to the support of the product of the integrands, which in this case is $[-\varepsilon, -0.3\varepsilon]$.

this approach to calculate $\varepsilon\sqrt{\cdot}$ are displayed in [figure 4\(b\)](#).

4 The mollified Kuwabara-Kono force model

Using the concepts introduced above, we propose a regularization of the Kuwabara-Kono model obtained by substituting the $\sqrt{\cdot}$ function — originally present in [equation \(3\)](#) — by $\varepsilon\sqrt{\cdot}$, as defined in [equation \(12\)](#). Note that there is nothing special about the $\sqrt{\cdot}$ function that allows it to be regularized via mollification; this means that this regularization approach can work for other non-smooth models. Combining this newly proposed model with the traditional Hertz model, the expression for the contact force between particles P_i and P_j now becomes

$$\|\mathbf{F}_i(t)\| = k_{i,j}\xi(t)^{3/2} + \gamma_\varepsilon\sqrt{\xi}\xi'(t). \quad (13)$$

With the removal of the singularity of the first derivative of $\varepsilon\sqrt{\xi}(t)$ at $t = 0$, the error of the method given in [equation \(4\)](#) should decrease as $\mathcal{O}(h^2)$.

4.1 Order for binary collisions

We performed the order-measuring procedure described in [subsection 2.3](#) using this new model, with the parameters given in [table 2](#). This procedure was repeated for different values of ε , in order to evaluate the effect that the mollification parameter has on the overall error behavior.

Table 2: Values of the parameters used in the order-measuring procedure (described in [subsection 2.3](#)) with the predictor-corrector leapfrog method applied to the mollified Hertz-Kuwabara-Kono force model.

Particle	Model	Initial Condition
$\rho = 19300\text{kg/m}^3$	$k_{1,2} \approx 4.4 \times 10^{10}\text{N/m}^{1.5}$	$v_0 = 1\text{m/s}$
$r = 1\text{m}$	$\gamma \approx 1.1 \times 10^9\text{kg}/(\text{m}^{0.5}\text{s})$	

The results are displayed in [figure 6](#) and indicate that the expected order of the method described in [equation \(4\)](#) — i.e. $\mathcal{O}(h^2)$ — was achieved when the model in [equation \(13\)](#) is used. In each of the plots in [figure 6](#), one observes the existence of three distinct regions. The boundary between these regions is highlighted in [figure 6\(a\)](#). For larger h , we observe a region where the error, although monotonically decreasing, does not have a clear order. Then, as h decreases, there is an intermediate region where the relative error oscillates. The length of this region depends on ε . Finally, there is a region, for smaller h , where the error becomes monotonically decreasing again, but now doing so as $\mathcal{O}(h^2)$.

The existence of these regions where the error fluctuates can be explained based on the discussion of [subsection 3.3](#) and, more specifically, by the observations made in [figure 3\(b\)](#), where it is shown that $\varepsilon\sqrt{x}$ and $\sqrt{x}$ are most different when x is close to zero. For values of h larger than about 2ε , the behavior of $\varepsilon\sqrt{x}$ near 0 is never relevant since, in the first step of the integration, the particle will already have crossed the region where $\varepsilon\sqrt{x}$ deviates the most from $\sqrt{x}$ and the duration of the simulation for the order analysis is too short for the particle to have time to bounce back. Thus, $\varepsilon\sqrt{x}$ is effectively very close to $\sqrt{x}$ and the model presented in [equation \(13\)](#) behaves almost as the original Hertz-Kuwabara-Kono model. This explains the behavior seen in the rightmost region of [figure 6\(a\)](#). However, as h becomes significantly smaller than 2ε , i.e. the leftmost region in [figure 6\(a\)](#), the collision of the particles is very well resolved, that is, there will be many time steps in the region of $\varepsilon\sqrt{x}$ near 0. In this

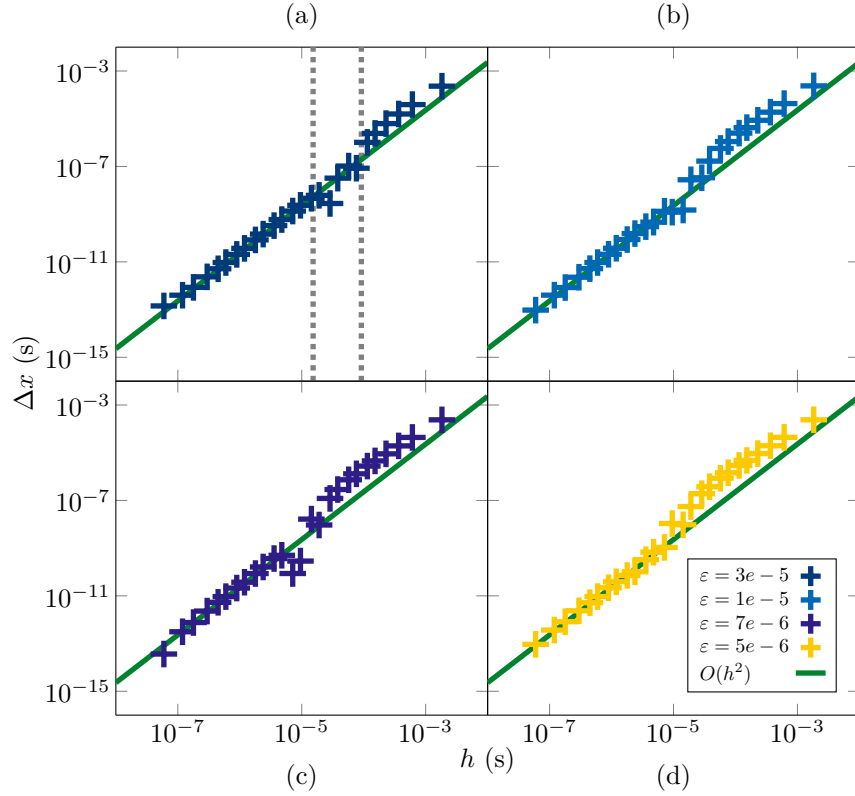


Figure 6: Results of the order-measuring procedure (described in subsection 2.3) with the predictor-corrector leapfrog method applied to the mollified Hertz-Kuwabara-Kono force model. The values of the parameters are presented in table 2.

left-most region, the $\mathcal{O}(h^2)$ convergence as $h \rightarrow 0$ is observed. Finally, for intermediary values of h , the integration of the forces will not sample $\varepsilon\sqrt{x}$ near 0 enough times, which causes the erratic behavior observed in the middle region of figure 6(a).

Therefore, in order to effectively use the regularized model proposed in equation (13), it is necessary that the value of h belongs to the leftmost region of the plots in figure 6. However, as the discussion in the last paragraph suggests, the exact position of the leftmost region is dependent on the value of ε , i.e., there exists a dependency of ε on h if one wishes to cross the threshold to $\mathcal{O}(h^2)$. In particular, we have observed empirically that this relation between h and ε is itself dependent on the other parameters of the simulation, e.g. particle radius and stiffness coefficient. Nonetheless, as a general rule, $h < \varepsilon$ is a necessary condition for $\mathcal{O}(h^2)$.

4.2 Physics of binary collisions

One can expect that regularizing the force scheme will affect the physical behavior of the contact. In order to evaluate the effect of the regularized model proposed in equation (13) on the physics of the problem, we performed some simulations to determine the influence of the regularization of the normal force, in particular the effects of the choice of the ε parameter of equation (13). We also performed qualitative comparisons of the coefficient of restitution of binary collision using the model described by equation (13) with the numerical results obtained with equation (3) presented in [25]. The parameters for the simulations performed in this sections are presented in table 3, which are identical to the ones used in [25].

Table 3: Values of the parameters used to perform the simulations used for validation of physical behavior of the binary collisions.

Particles	Normal Forces	Simulation
$\rho = 1300\text{kg/m}^3$	$\tilde{k}_n \approx 9 \times 10^7 \text{N/m}^{1.5}$	$h = 2^{-23}\text{s}$
$r = 3\text{mm}$	$\gamma = 190\text{kg/m}^{0.5}\text{s}$	$(\approx 1.19 \times 10^{-7}\text{s})$

We performed many simulations with setup almost identical to the one described in subsection 2.3, but with the contact force now being described by equation (13). Each of these simulations used a different value of the parameter ε . The trajectory of the particle with non-zero initial velocity was recorded and is displayed in figure 7. We observe that the trajectories of the particle obtained from the mollified model are very similar to the original model. There are changes in the duration of the collision, which are very minor, and in the maximum overlap, which are more noticeable. This change in maximum overlap increases as ε increases. Therefore, the first conclusion is that the regularized force model slightly softens the materials, allowing for slightly larger maximum overlaps during the collisions.

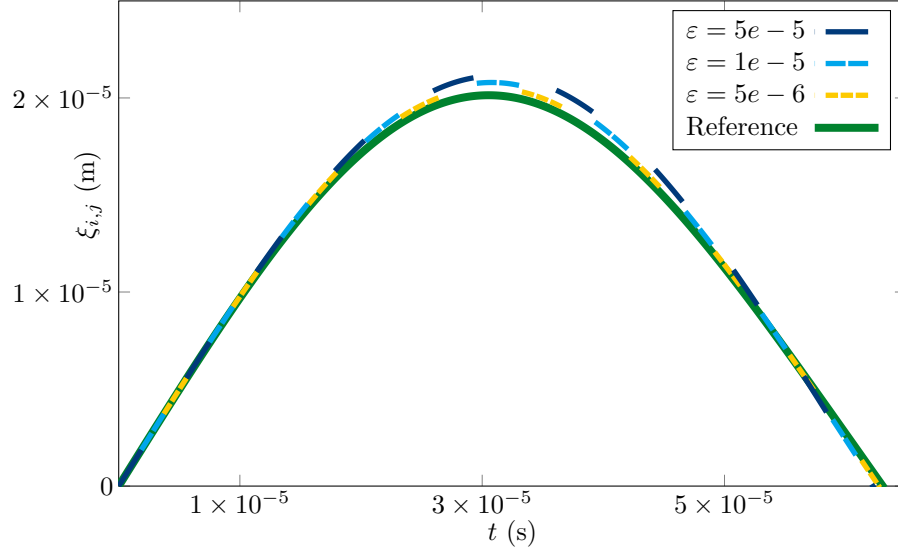


Figure 7: Time evolution of the position of the mobile particle in a normal binary collision using the regularized model in [equation \(13\)](#), for different values of ε . The reference curve instead uses the force model given in [equation \(3\)](#). The impact velocity used is 1m/s and the values of the remaining parameters are presented in [table 3](#).

The standard Hertz-Kuwabara-Kono force model produces coefficients of normal restitution, defined as

$$e_n = \frac{|\mathbf{v}_{\text{after}}|}{|\mathbf{v}_{\text{before}}|}, \quad (14)$$

that depend on the normal impact velocity [\[25\]](#). This dependence cannot be easily inferred from [figure 7](#). Thus, simulations were performed, in a manner much like the one described in [subsection 2.3](#), in which the coefficient of normal restitution was measured for different impact velocities and for different values of ε . The results can be seen in [figure 8](#).

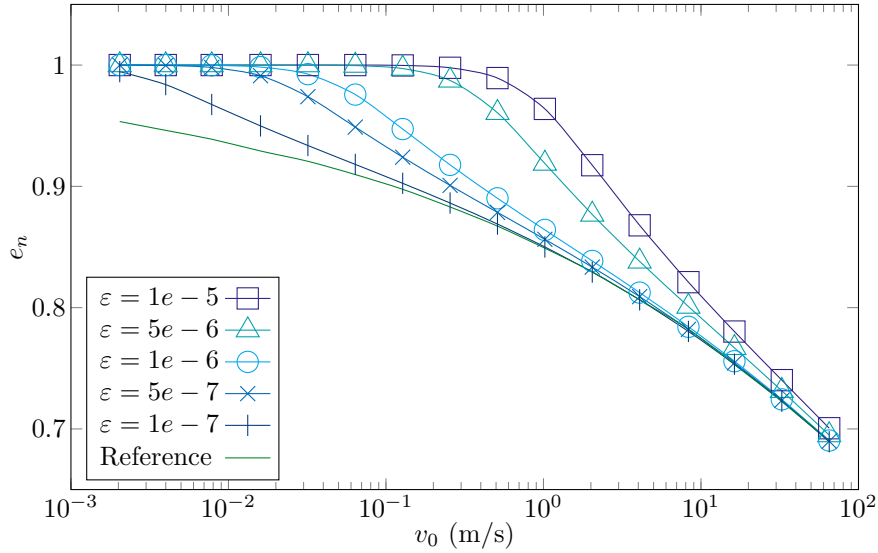


Figure 8: Visualization of the dependence of the coefficient of normal restitution on the impact velocity, for different values of ε . The values of the parameters can be found in [table 3](#). Reference curve obtained from [\[25\]](#).

As shown in [figure 8](#), collisions become more elastic (e_n closer to 1) as the value of ε increases, especially for very low normal impact velocities. The maximum overestimation of the restitution coefficient with respect to the reference value is of about 17% for normal impact velocities just under 1 meter per second, when $\varepsilon = 10^{-5}$. This effect, however, becomes significantly less pronounced as the normal impact velocities increase, especially as smaller values of ε can be chosen. In conclusion, from [figure 8](#) we observe that the regularized model in [equation \(13\)](#) produces slightly more elastic responses and hinders the efficiency of the normal dissipative terms during binary collisions.

Nevertheless, the results above indicate that the regularized model in [equation \(13\)](#) can be used as a viable alternative to model normal forces in DEM simulations. The differences observed with respect to the usual Hertz-Kuwabara-Kono model, in [equation \(3\)](#), are not too relevant if one considers the large variety and spread in the experimental data obtained for normal

impacts of particles [26]: although the Kuwabara-Kono model is one of the most widely used model for normal forces in DEM, it does not reproduce precisely the trends for all kinds of particles and materials.

5 Multiparticle flows

So far, we have only studied convergence of binary collisions. However, most applications of DEM in the field of granular media deal with flows involving thousands of grains or more. Thus, we need to assess the performance of the regularized model when used to simulate multiparticle flows.

However, attempting to measure the order of convergence of a multiparticle flow by recording some quantity — say, the x coordinate of the position on a particle — at the end of many simulations performed with different time step sizes, one is bound to be shocked. This is because such granular flows, chaotic by nature, rarely, if ever, have a well-defined notion of convergence, let alone of order.

An example of this can be found in section 5.1 of [4], where the same initial condition of a granular collapse was evolved with the same integration method but with several time step sizes. By tracking the motion of some individual particles in that example, one could see almost no sign of convergence of the trajectory or final position of the particles. Indeed, sometimes particles whose simulations had close time step sizes end up farther apart than the particles in simulations with a much larger gap in the time step sizes. This chaotic nature of multiparticle flows requires a new way to compare trajectories, which is the subject of the next subsection.

5.1 A metric for the comparison of simulations

In order to obtain a more complete description of the convergence of particle trajectories in an entire simulation, we introduce a metric which measures, for simulations that differ only in the choice of the step size h , the least upper bound on the distance of the trajectories of corresponding particles as a function of time.

More formally, let $N \in \mathbb{N}$ and suppose we run N simulations. For each $i \in \mathbb{N} \cap [1, N]$, there is a simulation, which we will denote by H_i , that is performed with some unique time step size $h_i \in (0, \infty)$, but uses same parameters and initial conditions as the others. Without loss of generality, we can assume $h_1 > h_2 > \dots > h_N$. Then, we must also assume that there exists $K_i \in \mathbb{N}$ such that say $K_i h_i = h_1$, for each $i \in \mathbb{N} \cap [1, N]$.

Let $T \in \mathbb{R}$ be the desired physical time of duration of the simulation (which must be chosen such that $(T/h_N) \in \mathbb{N}$) and define

$$\mathbb{T} := \left\{ nh_1 \in \mathbb{R} \mid n \in \mathbb{N} \text{ and } nh_1 \leq T \right\}. \quad (15)$$

Note that $\mathbb{T}$ is precisely the set of all moments in time that are “reachable” by all simulations, i.e. there exists $k_2, \dots, k_N \in \mathbb{N}$ such that $k_2 h_2 = \dots = k_N h_N \in \mathbb{T}$. In particular, if, for all $i \in \mathbb{N} \cap [1, N]$, we had that $K h_i = h_1 \implies K \notin \mathbb{N}$, we would have $\mathbb{T} = \{0\}$.

Now, let $M \in \mathbb{N}$ be the number of particles in one of these simulations (and, therefore, all simulations, since they all have the same initial condition) and $x_{i,j} : \mathbb{T} \rightarrow \mathbb{R}^2$ be the function that ascribes a 2-dimensional position to the particle $j \in \{1, 2, \dots, M\}$ in the simulation i at each moment in time $t \in \mathbb{T}$. Then, we define, for $t \in \mathbb{T}$,

$$d(H_1, \dots, H_N)(t) := \max_{1 \leq i \leq M} \max_{1 \leq j \neq k \leq N} \|x_{i,j}(t) - x_{i,k}(t)\|. \quad (16)$$

When the particles all have the same radius, this metric can also be nondimensionalized via the formula

$$\bar{d}(H_1, \dots, H_N)(\bar{t}) := \frac{1}{2r} d(H_1, \dots, H_N)(\bar{t}T), \quad (17)$$

where $r \in \mathbb{R}$ is such radius, $\bar{t} \in [0, 1]$ is the dimensionless time and T is as above.

Intuitively, the meaning of $\bar{d}(H_1, \dots, H_N)(\bar{t}) < 1$ is that, at time $\bar{t}$, the distance of the centers of any corresponding particles in any two of the N simulations — that is, particles that had the same initial position — is less than their diameter, i.e. all N simulations agree, at time $\bar{t}$, on the position of all particles up to a diameter in error. Although not mathematically rigorous, one would expect the true solution for the position of one of the particles to lie among some interpolation of the positions of the corresponding particles in the N simulations, as long as h_1 is taken small enough. This would, heuristically, mean that, if H_0 is a fictitious simulation that is identical to the true solution of the problem for all $t \in \mathbb{T}$, then $\bar{d}(H_0, H_k)(\bar{t}) < 1$, for any $k \in \mathbb{N} \cap [1, N]$.

5.2 Flow description

As depicted in figure 8, the behavior of the regularized model becomes closer to the original model as the impact velocities become larger. Because of this, we opted to use a granular explosion [23] as the setting for the comparison of the regularized and original model with regards to the evolution of the metric proposed in equation (17). For the purposes of this comparison, it is enough to simulate simplified versions of granular explosions, whose assembly we describe in the following.

First, a circle of radius 0.45m is filled by a maximally packed lattice of particles. This lattice is axis-aligned, with the rows being parallel to the x axis, while the columns form a zigzag pattern. There is no initial interpenetration among particles in this lattice.

If we consider the center of the circle to be on the axes’ origins (it can be anywhere, the flow is invariant by translations), we discard the particles whose centers are outside the first quadrant. We then place walls of fixed particles (i.e., particles whose motion will not be integrated during the simulation, but that can still produce contact forces) along the bottom and the left-side of this semicircle of particles, in such a way that the convex hull of the particles in the semicircle and each of the convex hulls

of the particles in the walls touch without intersecting. Each of these walls is 1000 particles long. This is done to restrict the space in which particles can move and, consequently, the amount of memory allocated to the spatial hash algorithm.

The particles in the walls and the ones in the outer part of the semicircle (specifically, their center are further away from the center of the circle than 0.225m) have their initial velocities set to 0. The other particles, i.e., the ones whose center are within 0.225m from the center of the semicircle, have a velocity vector with magnitude of 100m/s with an angle that is a slight random perturbation of the angle from the particle's center to the center of the semicircle. The choice of magnitude of 100m/s was made such that if the total kinetic energy of the system was evenly spread among all movable particles, the speed of each particle would be above 10m/s, which puts the relative error in the coefficient of normal restitution between the regularized and standard model well below 1%, as seen in [figure 8](#). The remaining properties of the particles and assembly can be found in [table 4](#). The initial condition and time evolution of the granular explosion are depicted in [figure 9](#).

Table 4: Values of the parameters of the simulations used to assess the performance of the regularized model for multiparticle flows.

Name	Value
Number of movable particles	1790
Physical time of simulation	5×10^{-4} s
Particle radius	5×10^{-3} m
Density	2650kg/m ³
Normal elastic constant (Hertz)	$\approx 1.76 \times 10^7$ N/m ^{1.5}
Normal damping constant (mollified Kuwabara-Kono)	50kg/(m ^{0.5} s)

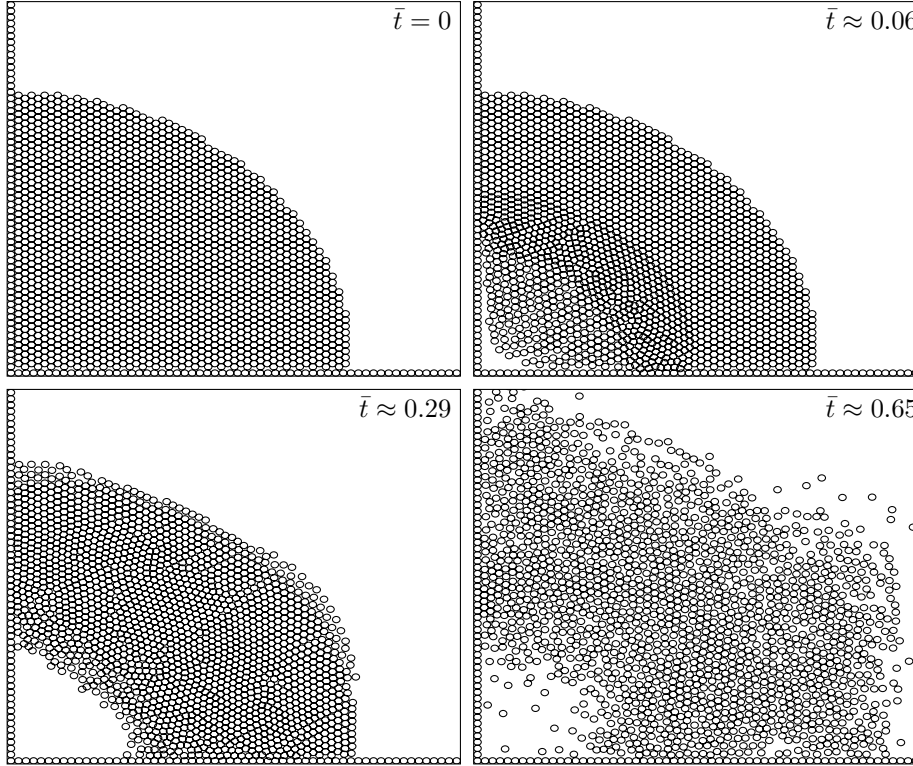


Figure 9: Time evolution of the simulation of the granular explosion described in [subsection 5.2](#), for the predictor-corrector leapfrog method along with the regularized force model of [equation \(13\)](#) and time step $h = 2^{-33}$. In the top-right of each image is the normalized time in which the snapshot was taken.

5.3 Results

An initial condition was generated according to [subsection 5.2](#) and this exact same initial condition was used in all subsequent simulations described in this subsection. We ran four different combinations of integration method and force model:

(PCL+MKK) The predictor-corrector leapfrog method ([equation \(4\)](#)) along with the new mollified Hertz-Kuwabara-Kono force model presented in [equation \(13\)](#).

(PCL+KK) The predictor-corrector leapfrog method (equation (4)) with the standard Hertz-Kuwabara-Kono force model presented in equation (3).

(SE+MKK) The symplectic (also known as semi-implicit) Euler method [27] along with the new Hertz-Kuwabara-Kono mollified force model presented in equation (13).

(SE+KK) The symplectic Euler method [27] along with the new Hertz-Kuwabara-Kono force model presented in equation (3).

The first two items are used for a direct comparison between the new model presented in equation (13) and the standard Hertz-Kuwabara-Kono model. In order to verify that any improvement in (PCL+MKK) in relation to (PCL+KK) are solely due to the increase in order caused by the regularized model, we also ran (SE+MKK) and (SE+KK) as control simulations.

For each of the four combinations of integration method and force model, we ran nine simulations with different time step sizes. They range from $h = 2^{-25} \approx 2.98 \times 10^{-8} \text{ s}$ to $h = 2^{-33} \approx 1.16 \times 10^{-10} \text{ s}$, with each subsequent value being half of the previous. The value of the largest time step size was chosen so that even the collisions of the fastest moving particles in the assembly would last at least 100 steps. The choice of using powers of 2 is to remove any errors from floating point representations, to guarantee a logarithmic spacing of the time steps of the simulations and to make sure that the elements of the set $\mathbb{T}$ are not too spaced out.

Whenever the mollified model was involved, we used $\varepsilon = 10^{-5}$. This value was chosen because it was close to the highest such that binary collisions of particles using the parameters in table 4, with speeds of up to 100m/s would still safely yield order two when integrated with the predictor-corrector leapfrog method. We maintained the value of ε constant among all simulations using the mollified model, so that they all solve the same ODE system. We remind the reader that, in practice, as one reduces the time step size h , one can safely reduce the value of ε to get a regularized model closer to the Hertz-Kuwabara-Kono model while still maintaining $\mathcal{O}(h^2)$ convergence.

The nondimensionalized time evolution of the metric $\bar{d}$ applied to the nine simulations of each integration method and force model combination is presented in figure 10. Since the symplectic Euler method is first order accurate, we do not expect to see any improvement in convergence gained by regularizing the method. This is clearly visible in figure 10, because both models integrated with the symplectic Euler method had about the same time evolution of $\bar{d}$. Furthermore, the fact that their evolution matches so closely is a good indicator that, for this value of ε , the mollified model is a good approximation of the standard model. For the simulations integrated with the predictor-corrector leapfrog, figure 10 shows that, at any specific point in nondimensionalized time, the simulations which used the standard model have deviated amongst themselves about one order of magnitude (in nondimensionalized distance) more than the ones which used the regularized model. This indicates that, at least in this type of multiparticle flow, the regularized contact model given by equation (13) works in keeping particle trajectories closer to the true solution.

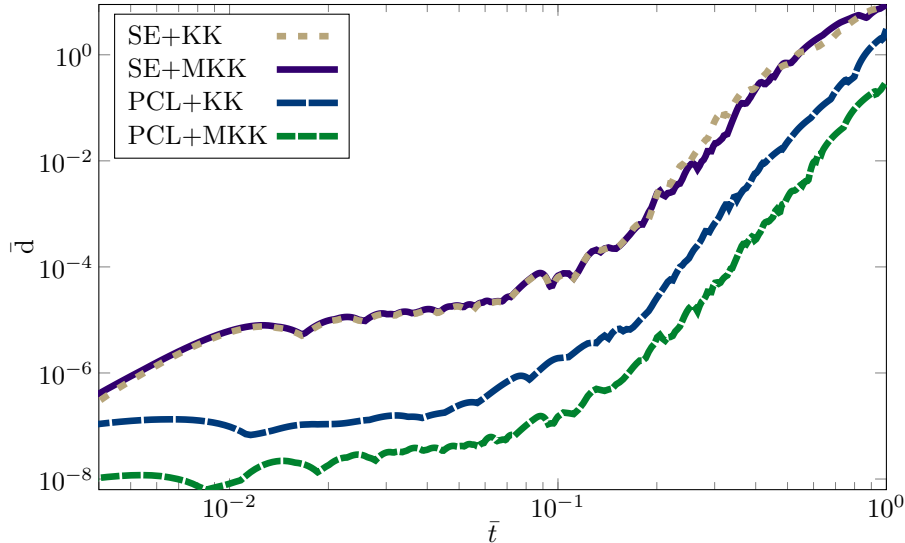


Figure 10: Nondimensionalized time evolution of $\bar{d}$ (given in equation (17)) applied to nine simulations with $h \in \{2^{-25}, 2^{-26}, 2^{-27}, 2^{-28}, 2^{-29}, 2^{-30}, 2^{-31}, 2^{-32}, 2^{-33}\}$, for four combinations of integration method and force model. The setup of the simulations is described in subsection 5.2 and further values of the parameters can be found in table 4. Whenever applicable, the value of the parameter ε used in the regularized model of equation (13) is 10^{-5} .

6 Conclusion

In this work, we have used the mollification process to regularize the square root function that appears as part of the Kuwabara-Kono force model. Mathematically, the proposed regularized model fully solves the order defect described in [5,

18], and one can tune the model arbitrarily close to the standard one by making $\varepsilon \rightarrow 0$. However, the dependence of the time step size on ε in order to achieve the $\mathcal{O}(h^2)$ convergence imparts an increase in computational cost.

We have shown, in practice, that binary collisions using the regularized model do converge with order two. This regularized force model, however, mimics slightly softer materials and slightly less dissipative collisions with respect to a similar simulation carried out for the same parameters using the Kuwabara-Kono model. Still, for most real cases, this is within the range of fluctuations of experimental measurements in real particles.

We have also displayed an example of a multiparticle flow where the improved order provided by the regularization of the model reduces the deviation among numerical solutions, when compared to the same flow using the standard Hertz-Kuwabara-Kono model instead. The parameters used in this multiparticle flow are consistent with real use cases; in particular, the time step size $h \approx 2.98 \times 10^{-8}$ is still within computational feasibility. For this choice of parameters and $\varepsilon = 10^{-5}$, the regularized model appears to approximate well the standard model.

More work is necessary in order to fully characterize the regularized model. In particular, a parameter study for the coefficient of restitution and collision duration still needs to be performed. Additionally, further research is required as to alleviate the dependence of the time step size on ε .

Beyond its application in granular materials, this mollification approach can be used to numerically approximate the solution to other ODEs whose right-hand side is non-Lipschitz near the initial condition, and it can do so with arbitrarily high order and fidelity to the original problem.

Finally, our current direction of research is related to fixing a similar order defect that happens when tangential forces are included via the Cundall-Strack friction. Assuming compatibility, this would allow one to fully integrate viscous and frictional systems with second order accuracy, which is the long term goal of our research.

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